UNIVERSITY OF MAURITIUS
SIS 3204Y(5) – NATURAL LANGUAGE PROCESSING

MOCK PAPER 4 – Weeks 1 to 8 (Question Paper with Model Answers)
Total: 50 marks | Suggested duration: 60 minutes

Instructions: Answer ALL questions. Marks are shown beside each question. The week each question comes from is shown in the question header. Model answers are given under each question in bullet points; the mark for a point is shown in brackets, e.g. (1). Cover the answers while practising. Equivalent correct answers should receive credit.

Question 1 | Week 1 (Revision Q12) [3 marks]

What is hallucination in a Large Language Model?

Answer

- **Definition:** the LLM generates output that is factually incorrect, ungrounded or completely made up. (1)
- The text often sounds authoritative, logical and convincing even though it is false. (1)
- Example: inventing non-existent academic references, historical facts or technical statistics. (1)

Question 2 | Week 2 (Revision Q13) [4 marks]

Explain why lowercasing improves NLP processing, using the words “NLP”, “Nlp” and “nlp” as an example. State one case where lowercasing can cause problems.

Answer

- Without lowercasing, “NLP”, “Nlp” and “nlp” are treated as **different words**; with lowercasing they all become “nlp”. (1)
- Benefits: smaller vocabulary, better matching and retrieval, and better generalisation of models. (2)
- Problem case: **Named Entity Recognition (NER)**, because capital letters signal names (proper nouns). (1)

Question 3 | Week 3 (Revision Q10 and Q8) [10 marks]

(a) Compare sparse and dense representations using a table with SIX aspects. (6 marks)

(b) What is the sparse vector problem? A model uses a vocabulary of 100,000 words. Calculate the number of entries needed to store 1 million one-hot vectors and the percentage that are zeros. (4 marks)

Answer

(a) Sparse (One-Hot, BoW, TF-IDF) vs Dense (Word2Vec, GloVe, FastText, BERT)

- **Vector size:** sparse = very high (10K to 1M+, equal to vocabulary size); dense = low (typically 50–500). (1)
- **Non-zero values:** sparse = very few (mostly 0s); dense = all or almost all non-zero. (1)
- **Data type:** sparse = binary (0/1), counts or TF-IDF; dense = real-valued numbers. (1)
- **Captures meaning:** sparse = no; dense = yes. (1)
- **Captures similarity:** sparse = no (all words equally distant); dense = yes (similar words are close). (1)
- **Memory / computation:** sparse = high memory, slower; dense = low memory, faster. (1)
- Key idea: sparse vectors tell us *which* words are present; dense vectors tell us *what words mean* and how they relate.

(b)

- **Sparse vector problem:** one-hot vectors are mostly 0s with a single 1, so for large vocabularies they become huge and mostly uninformative. (1)
- Each vector has length 100,000, so 1,000,000 × 100,000 = **100 billion entries**. (1)
- Only 1 million entries are 1s, so **99.999%** of the entries are zeros. (1)
- Consequences: high memory use, wasted computation on zeros, inefficient storage, not scalable. (1)

Question 4 | Week 4 (Part C Q9) [4 marks]

Loss: The model gives probability 0.8 to the correct class. Calculate the loss $L = -\ln(0.8)$. If P(Positive) increases to 0.95, what happens to the loss? Explain.

Answer

- Formula and substitution: $L = -\ln(0.8)$. (1)
- $L \approx$ **0.223**. (1)
- With P = 0.95 the loss becomes **smaller** (about 0.051). (1)
- Because the model assigns a **greater probability to the correct class**. (1)

Question 5 | Week 5 (Sample Q8) [10 marks]

You are developing a system to classify long customer reviews as positive, neutral or negative. Some reviews contain important information near the beginning that affects the interpretation of words much later in the text. Discuss whether an RNN, LSTM or GRU would be appropriate for this task.

Answer

- **Requirement:** the system must keep information across many time steps (long-term dependency). (1)
- **RNN:** can do sentiment classification, but vanishing gradients make it hard to learn long-term dependencies, so it is not the best choice for long reviews. (2)
- **LSTM suitable:** designed to keep information for longer using the cell state and the forget, input and output gates. (2)
- **LSTM cost:** more parameters and greater computational cost. (1)
- **GRU suitable:** gated memory with reset and update gates; can still model long-term dependencies. (2)
- **GRU efficiency:** simpler than LSTM and normally has fewer parameters. (1)
- **Model selection:** evaluate both LSTM and GRU, and choose using computing resources and validation performance. (1)

Question 6 | Week 6 (Set 1 Q9) [10 marks]

What is multi-head attention and why can it be more useful than using only one attention head?

Answer

- Several attention operations (heads) run **in parallel**. (2)
- Each head uses **different learned projections** of Q, K and V. (2)
- Each head produces its own output. (1)
- The head outputs are **concatenated** and passed through an **output projection**. (2)
- **Benefit:** the model can learn multiple attention patterns and representation subspaces in the same layer. (2)
- **Caution:** a head should not automatically be assumed to match one fixed linguistic rule. (1)

Question 7 | Week 7 (Set 2 Q4) [5 marks]

A GPT model is prompted with “The meeting starts at”. Its next-token probabilities are: nine 0.45; ten 0.35; noon 0.15; tomorrow 0.05.

a) Which token will greedy decoding select? (1 mark)

b) Explain why sampling might produce a different token from greedy decoding. (2 marks)

c) Describe the two steps GPT repeats to produce a longer response after selecting its first token. (2 marks)

Answer

(a)

- **“nine”** – it has the highest probability (0.45). (1)

(b)

- Greedy decoding **always** picks the most probable token. (1)
- Sampling draws from the probability distribution, so it can choose another token, such as “ten”. (1)

(c)

- **Append** the selected token to the context. (1)
- Run **next-token prediction again** and repeat until a stop token or other stopping rule. (1)

Question 8 | Week 8 (Tutorial Set A Q1) [4 marks]

(a) Define a foundation model. (2 marks)

(b) Explain why not all foundation models are LLMs. Give one example. (2 marks)

Answer

(a)

- A foundation model is trained with **broad, large-scale pre-training**. (1)
- It can be **adapted for many downstream tasks**. (1)

(b)

- Foundation models can focus on other modalities (images, audio), while LLMs mainly process and generate **language**. (1)
- Example: a **vision foundation model** is not necessarily an LLM. (1)

Marks Summary

Question	Week	Source	Marks
1	1	Revision Q12	3
2	2	Revision Q13	4
3	3	Revision Q10 and Q8	10
4	4	Part C Q9	4
5	5	Sample Q8	10
6	6	Set 1 Q9	10
7	7	Set 2 Q4	5
8	8	Tutorial Set A Q1	4
		TOTAL	**50**